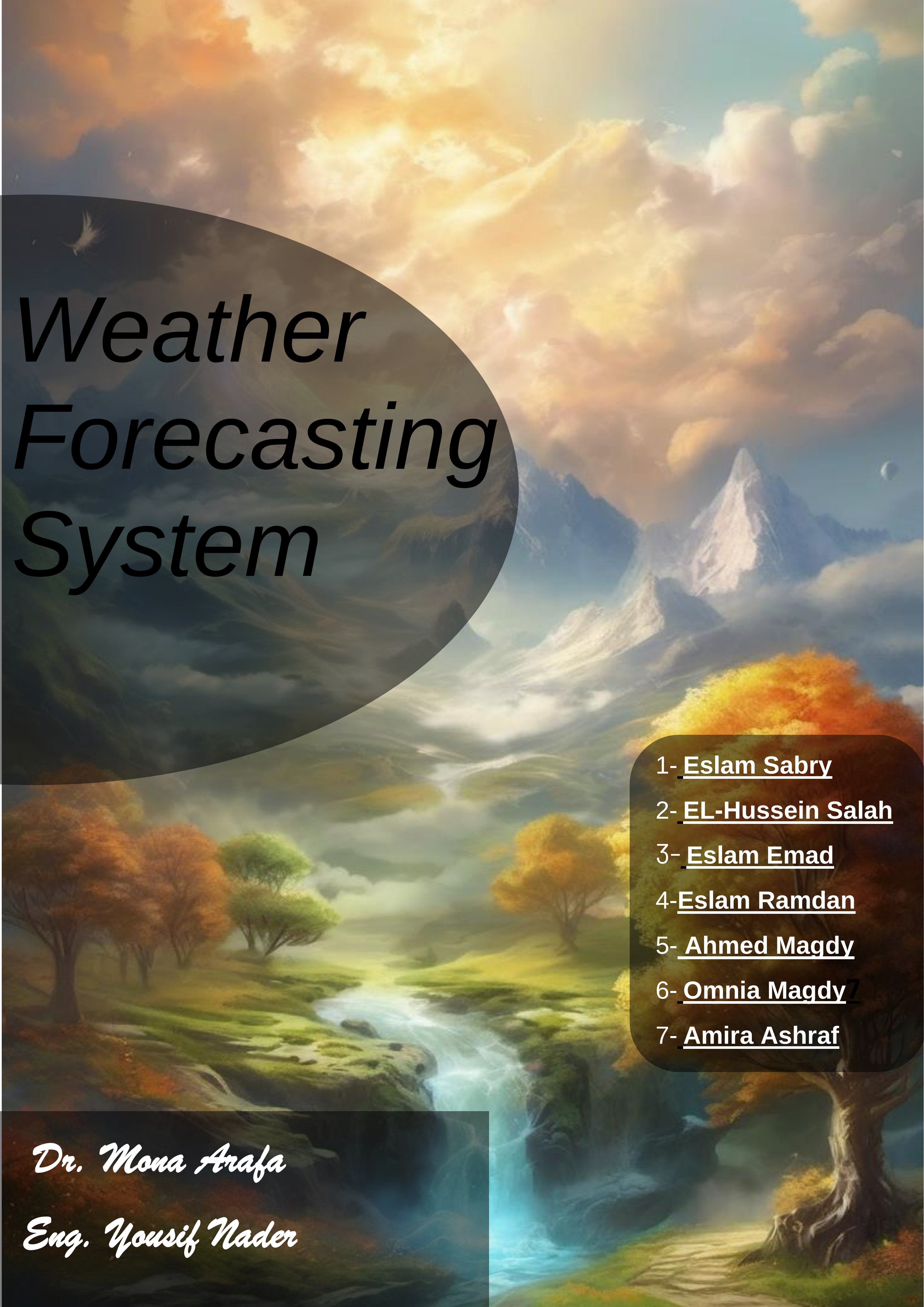

Weather Forecasting System

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Weather Forecasting System

Contents

Section 1 Project Overview

1.1 Project Description

The project involves the design, development, and implementation of a weather forecasting system that utilizes various data sources such as satellite imagery, weather sensors, historical data, and atmospheric models. The system will employ machine learning and statistical techniques to analyze and interpret this data, generating forecasts for different regions and timeframes.

Project Objectives:

- Develop a robust and scalable software platform for collecting, processing, and analyzing weather data from diverse sources.
- Implement advanced forecasting algorithms and models to generate accurate and reliable weather predictions for various timeframes.
- Design intuitive user interfaces for accessing weather forecasts, incorporating interactive features and customizable settings.
- Enhance system performance and responsiveness to ensure timely delivery of weather updates and alerts.
- Provide comprehensive documentation, training resources, and user support to facilitate the effective use of the system.

1.2 Project Scope

- Gathering and processing weather data from multiple sources.
- Developing algorithms and models for weather prediction.
- Building a user-friendly interface for accessing weather forecasts.
- Testing and validating the accuracy of the forecasting system.
- Deploying the system to serve end-users, which may include individuals,
- businesses, and government agencies.

Project Includes
Data Acquisition
Data Processing and Analysis
Visualization and Presentation
Alerting and Notification
Geospatial Analysis
Integration with External Systems
Historical Data Analysis

Project Excludes
Long-Term Climate Modeling
Specialized Meteorological Research
Hardware Development
Proprietary Data Sources
Legal and Regulatory Compliance

1.3 Assumptions

Assumptions
Sufficient data availability: The project assumes that there will be access to a diverse range of weather data sources, including satellite imagery, sensors, and historical records.
Availability of expertise: It is assumed that the project team will have access to experts in meteorology, data science, and software development to successfully execute the project
Support for infrastructure: The project assumes that the necessary infrastructure, such as computing resources and network connectivity, will be available to support the development and deployment of the forecasting system.

1.4 Constraints

Constraints
• **Technological Constraints:** ◦ Compatibility with existing infrastructure and systems may impose technological constraints. ◦ Limited access to advanced technology or computing resources may affect the scalability and performance of the system.
• **Budget Constraints:** ◦ Limited budget allocation for the project may constrain the scope of development and resources available.

Constraints
• Budget constraints may impact the selection of tools, technologies, and resources for the project.
• **Time Constraints:** • Tight project timelines and deadlines may limit the amount of time available for development, testing, and deployment. • Seasonal variations or specific event-related deadlines (e.g., hurricane season) may impose time constraints on project delivery.
• **Regulatory Constraints:** • Compliance with regulatory requirements and standards related to data privacy, security, and accessibility may constrain system development. • Adherence to weather data licensing agreements and regulations may restrict data usage and distribution
• **Resource Constraints:** • Limited availability of skilled personnel with expertise in meteorology, data science, and software development may constrain resource allocation. • Constraints on equipment, software licenses, and facilities may impact the implementation and deployment of the system.
• **Stakeholder Expectations:** • Conflicting stakeholder expectations and requirements may impose constraints on project scope and deliverables. • Balancing the needs of diverse stakeholders (e.g., general public, businesses, government agencies) may introduce constraints on system design and functionality.
• **Scope Constraints:** • Defined project scope and objectives may limit the extent of functionalities and features that can be included in the system. • Changes in project scope during development may introduce constraints on resources, timeline, and budget.
• **External Dependencies:**

Constraints
• Dependencies on external data sources, APIs, or third-party services may introduce constraints on system integration and reliability. • Unforeseen disruptions or changes in external dependencies may impact project timelines and deliverables.

Section 2 Project Start-Up

2.1 Project Life Cycle

Phase	Activities	Sequences
Planning	• Establishing the project initiation team • Establishing a relationship with the customer • Establishing the project initiation team plan • Establishing the project management environment and project workbook • Developing the project charter	Phase 1
Analysis	• Preliminary Investigation • The study of the current system (system survey) • General recommendation on how to fix, enhance and replace current system • Determination of the user requirements • Analysis of the system survey • Systems analysis reports	Phase 2
	• by taking the data which is collected on atmospheric conditions and record temperature, rainfall, humidity and other parameters.	

Design	• by using Cloud computing and Data mining techniques • Statistical framework is provided on Modeling for surface process of weather forecasting with Data assimilation • Proposed algorithm is used for the construction of decision tree. • report is implemented by using webApp.	Phase 3
Implementation **(coding - testing)**	• Construct software components • Verify and test • Train the users and document the systems • Install the system • Problems: • Inherent uncertainties • Limited forecast range • Accessibility and communication issues • Solutions: • Improved forecasting models • Probabilistic forecasting • Effective communication: • Public education and awareness • Collaboration and partnerships	Phase 4

2.2. Methods, Tools, and Techniques

1. **Development Methodologies:**
   - Agile Methodology: Iterative and incremental development approach, allowing for flexibility and adaptability to changing requirements.
   - Waterfall Methodology: Sequential development approach with distinct phases (e.g., requirements, design, implementation, testing), suitable for well-defined and stable requirements.
2. **Standards and Policies:**
   - ISO 9001: Quality management standards ensuring consistency and quality in project deliverables.
   - Data Quality Standards: Adherence to standards such as WMO (World Meteorological Organization) standards for weather data quality.
   - Accessibility Standards: Compliance with accessibility standards (e.g., WCAG) for ensuring accessibility of user interfaces.
3. **Procedures:**
   - Development Procedures: Standard operating procedures (SOPs) for software development, including coding standards, testing procedures, and documentation guidelines.
   - Deployment Procedures: Procedures for deploying software updates, managing configurations, and ensuring system reliability during deployment.
4. **Programming Languages:**
   - Python: For data processing, statistical analysis, and algorithm development.
   - Java: For building scalable and robust backend systems.
   - HTML and CSS: For building and style structure web application.
   - JavaScript: For developing interactive user interfaces and web-based applications.
   - R: For statistical analysis and modelling of weather data.
5. **Reusable Code Repositories:**
   - GitHub: Version control and collaboration platform for hosting code repositories, managing branches, and facilitating code reviews.
   - GitLab: Similar to GitHub, providing version control, CI/CD pipelines, and project management tools.
   - Bitbucket: Code hosting platform offering Git and Mercurial repositories, along with issue tracking and project management features.
6. **Notations:**
   - Unified Modeling Language (UML): Standard notation for visualizing, specifying, constructing, and documenting the artifacts of software systems.
   - Data Flow Diagrams (DFD): Notation for representing the flow of data within a system, illustrating how data is processed and transformed.
   - Entity-Relationship Diagrams (ERD): Notation for modeling the structure of databases, depicting entities, attributes, and relationships.
7. **Tools:**
   - Integrated Development Environments (IDEs): Such as PyCharm, IntelliJ IDEA, or Visual Studio Code for writing, debugging, and testing code.
   - Data Visualization Tools: Such as Matplotlib, Seaborn, or Plotly for visualizing weather data and forecast results.
   - Database Management Systems (DBMS): Such as PostgreSQL, MySQL, or SQLite for storing and managing weather data.
   - Geographic Information Systems (GIS): Such as ArcGIS, QGIS, or Leaflet for spatial analysis and visualization of weather data.

8. **Techniques:**
    - Continuous Integration/Continuous Deployment (CI/CD): Automation techniques for integrating code changes, running tests, and deploying updates to production environments.
    - Test-Driven Development (TDD): Development technique involving writing tests before writing code, ensuring code quality and test coverage.
    - Containerization: Using technologies like Docker or Kubernetes for packaging applications and dependencies into containers for easier deployment and scalability.

By utilizing these methods, tools, and techniques, the development and deployment of the Weather Forecasting System can be conducted efficiently and effectively, meeting project requirements and quality standards.

2.3 Estimation Methods and Estimates

Estimation Methods and Estimates	
Description	The estimation of project effort, schedule, and budget for the Weather Forecasting System project is based on a combination of expert judgment, historical data analysis, and bottom-up estimation techniques. **[Best / Most Likely / Worst]:** • **Best Estimate:** Represents the optimistic scenario where all tasks are completed efficiently without encountering significant delays or obstacles. • **Most Likely Estimate:** Represents the most realistic scenario based on expected productivity levels, potential risks, and uncertainties. • **Worst Estimate:** Represents the pessimistic scenario accounting for potential delays, setbacks, and unforeseen challenges.
Effort in person-months or person-hours	**Best Estimate:** 2000 person-hours

	Most Likely Estimate: 2500 person-hours **Worst Estimate:** 3000 person-hours
Schedule in calendar months	**Best Estimate:** 1.5 months **Most Likely Estimate:** 3 months **Worst Estimate:** 5 months
Budget in dollars	**Best Estimate:** $100,000 **Most Likely Estimate:** $125,000 **Worst Estimate:** $150,000
Level of Uncertainty	30%

2.4 Schedule Allocation

Major Milestone/Deliverable	Efforts
1. Establishing the project initiation team 2. Establishing a relationship with the customer 3. Establishing the project initiation team plan 4. Establishing the project management environment and project workbook	3 days 7 days 2 days 2 days

5. Preliminary Investigation 6. The study of the current system (system survey) 7. Determination of the user requirements Systems analysis reports	3 days 30 days 5 days
8. using Cloud computing and Data mining techniques 9. Proposed algorithm is used for the construction of decision tree. 10. Statistical framework is provided on Modeling for surface process of weather forecasting with Data assimilation	3 days 3 days 2 days
11. Probabilistic forecasting 12. Collaboration and partnerships 13. Construct software components 14. Verify and test 15. Train the users and document the systems 16. Install the system	12 days 7 days 2 days 7 days 10 days 2 days

2.5 Resource Allocation

Resource	Skill Set Requirements
Analyst	1- SQL 2- Data visualization 3- Critical thinking 4- Math and Statistics 5- communication
computer tool	- Pycharm - IntelliJ IDEA - Visual studio Code
Database designer	Design and integrate database
Designers	1. Technical ability of math, science and technology 2. Software skills such as Matlab, HMI, Simulink and PCI
Developers	Software language: Java, R, HTML, CSS, JS Python
Graphic tool	Illustrator
Inspectors	Testing and troubleshooting

Interface designer	UI design
Microsoft project	-The MSN Weather app
Project manager	Risk management, organization management and Communication skills
shared computer room	-
SQL server	Oracle server database
Trainers	Communication skills
training room	-

2.6 Budget Allocation

Activities	Budget
Establishing a relationship with the organization	9842.50 $
Establishing the project management environment and project workbook	10816.75 $
Developing the project charter	4921.25 $
The study of the current system	5905.50 $
Determination of the user requirements	13779.50 $
Analysis of the system survey	7874 $
Systems analysis reports	1968.50 $
by using Cloud computing and Data mining techniques	984.25 $
report is implemented by using Android.	787.40 $

Statistical framework of weather forecasting with Data assimilation	1181.10 $
Public education and awareness	5905.50 $
Construct software components	17706.50 $
Collaboration and partnerships	36392.25 $

Section 3. Risk Management

Risk Description	Probability	Impact	Strategy
Climate Change	**Low**	**Low**	Develop new techniques to predict extreme weather events.
Failure in Weather Prediction	**Low**	**High**	Improve the accuracy of prediction models and develop a backup system to address inaccuracies.
Impact of Artificial Intelligence Techniques	**Low**	**High**	Improve the accuracy of artificial intelligence systems and enhance transparency in models and analyses.
Impact of Data Absence or Delay	**High**	**High**	Improve sensor systems and ensure continuity of data provision.

Effect of Ineffective Communication	**High**	**High**	Enhance collaboration between relevant parties and adopt effective communication mechanisms.
Weather Variability	**High**	**High**	Implement ensemble forecasting techniques, invest in advanced modeling algorithms, and continuously validate and improve forecasting models.
Resource Constraints	**Medium**	**Medium**	Invest in training and development programs, explore outsourcing options for specialized tasks, and optimize resource utilization.
Time Constraints	**High**	**Medium**	Implement agile project management methodologies, prioritize tasks based on criticality, and plan for contingency buffers.

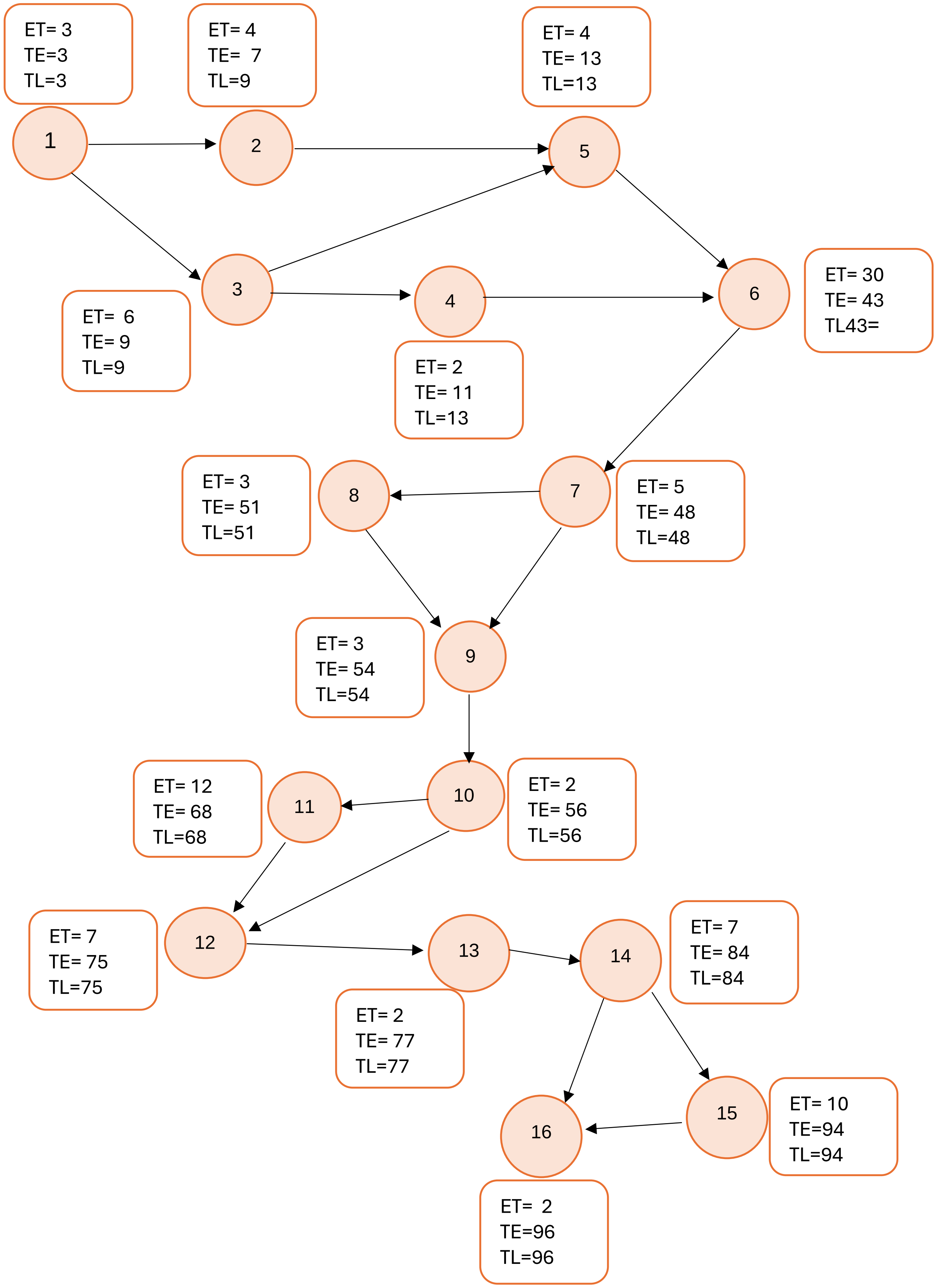
Activity No.	Activity	Time (Days)	Immediate predecessors
1	Establishing the project initiation team	3 days	------------
2	Establishing a relationship with the customer	7 days	1
3	Establishing the project initiation team plan	2 days	1
4	Establishing the project management environment and project workbook	2 days	3
5	Preliminary Investigation	3 days	2,3
6	The study of the current system (system survey)	30 days	4,5
7	Determination of the user requirements Systems analysis reports	5 days	6
8	using Cloud computing and Data mining techniques	3 days	7
9	Proposed algorithm is used for the construction of decision tree.	3 days	7,8
10	Statistical framework is provided on Modeling for surface process of weather forecasting with Data assimilation	2 days	9
11	Probabilistic forecasting	12 days	10

12	Collaboration and partnerships	7 days	10,11
13	Construct software components	2 days	12
14	Verify and test	7 days	13
15	Train the users and document the systems	10 days	14
16	Install the system	2 days	13,14

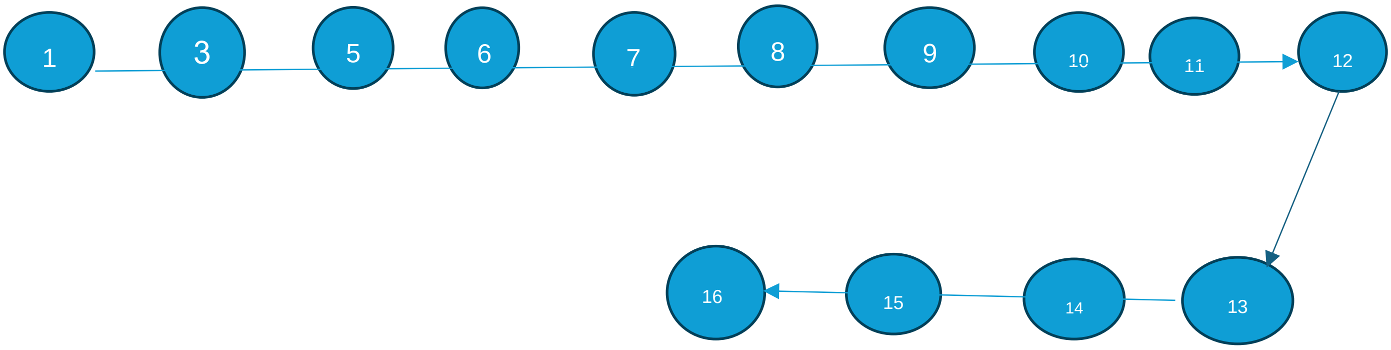
PERT (Program evaluation review techniques)

Activity	TE	TL	Slack (TL -TE)	ON Critical
1	3	3	0	√
2	10	10	0	√
3	5	11	**6**	×
4	7	13	**6**	×
5	13	13	0	√
6	43	43	0	√
7	48	48	0	√
8	51	51	0	√
9	54	54	0	√
10	56	56	0	√
11	68	68	0	√
12	75	75	0	√
13	77	77	0	√
14	84	84	0	√
15	94	94	0	√
16	96	96	0	√

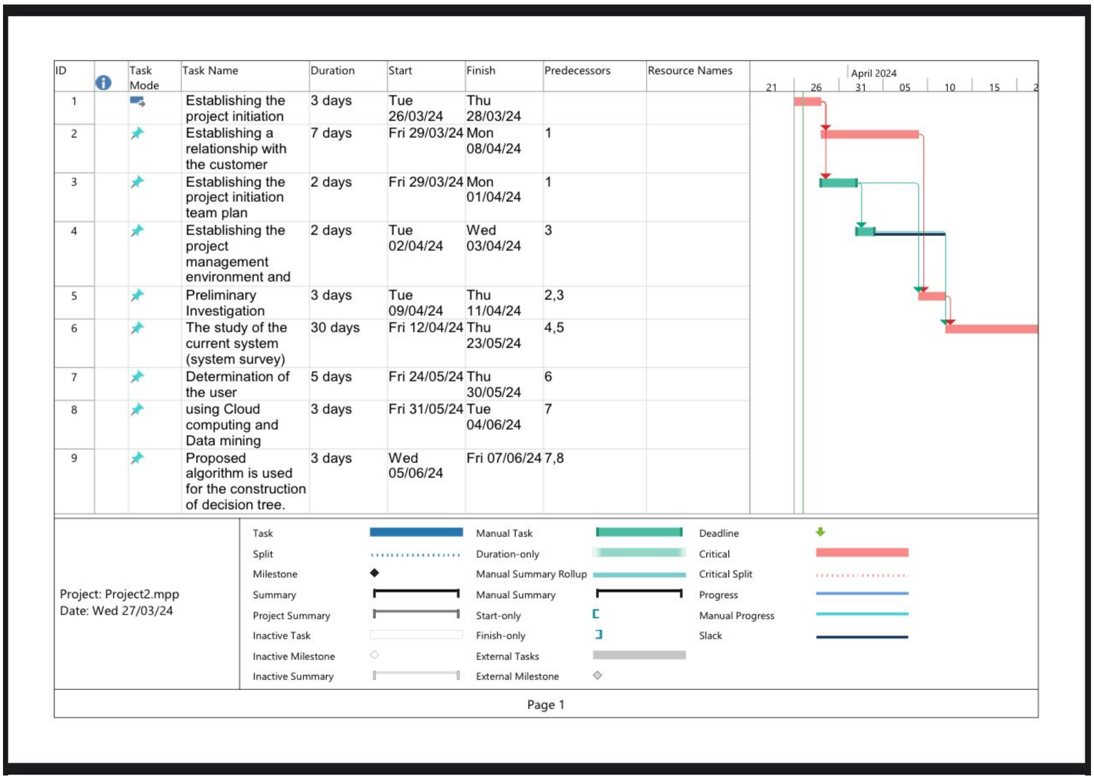
Network diagram:-



Critical path:



Gent:

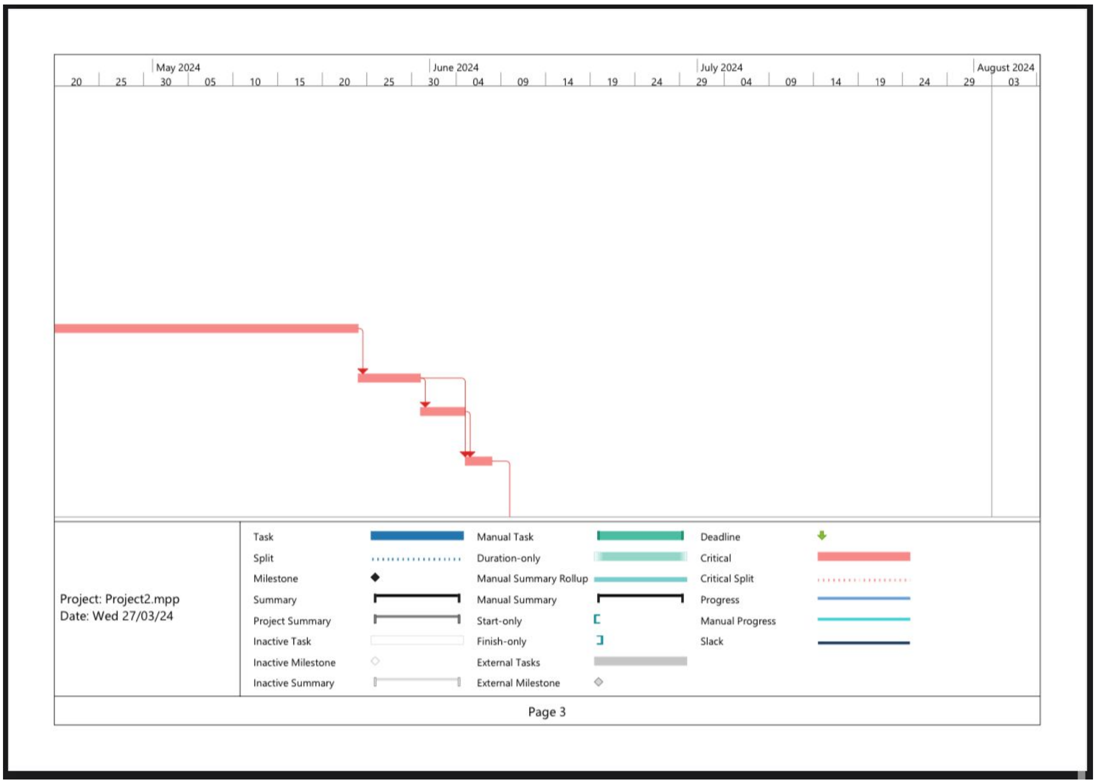


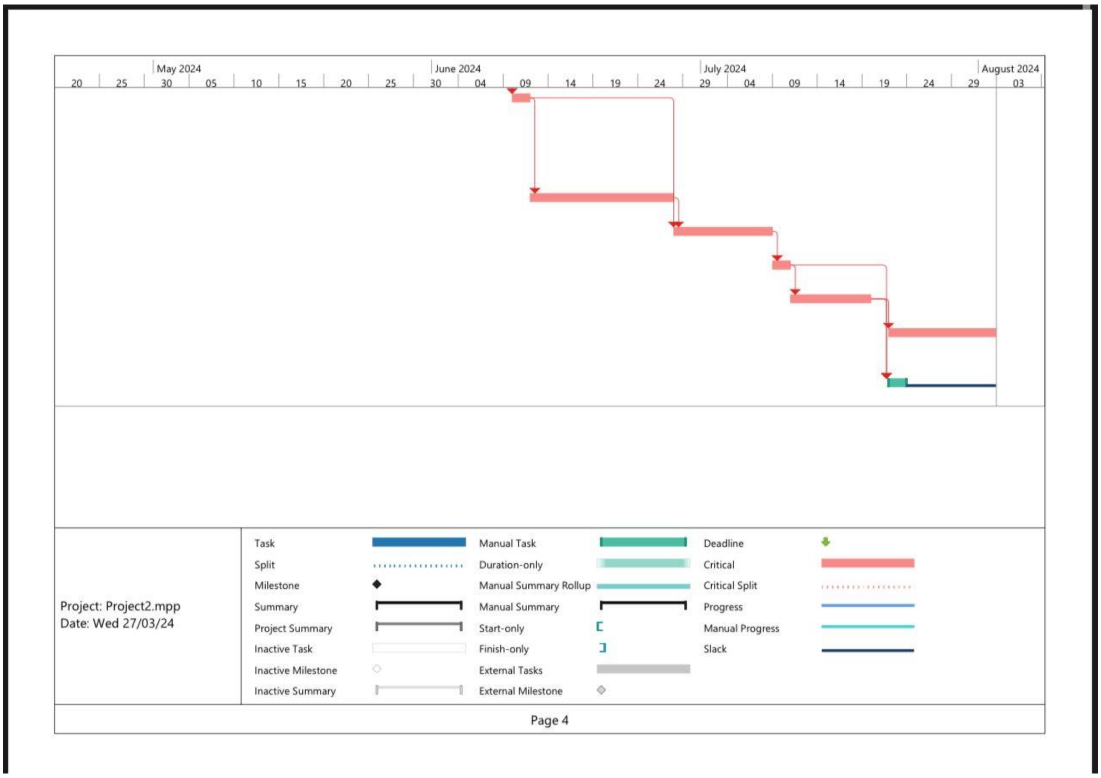
ID		Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names
1			Establishing the project initiation	3 days	Tue 26/03/24	Thu 28/03/24		
2			Establishing a relationship with the customer	7 days	Fri 29/03/24	Mon 08/04/24	1	
3			Establishing the project initiation team plan	2 days	Fri 29/03/24	Mon 01/04/24	1	
4			Establishing the project management environment and	2 days	Tue 02/04/24	Wed 03/04/24	3	
5			Preliminary Investigation	3 days	Tue 09/04/24	Thu 11/04/24	2,3	
6			The study of the current system (system survey)	30 days	Fri 12/04/24	Thu 23/05/24	4,5	
7			Determination of the user	5 days	Fri 24/05/24	Thu 30/05/24	6	
8			using Cloud computing and Data mining	3 days	Fri 31/05/24	Tue 04/06/24	7	
9			Proposed algorithm is used for the construction of decision tree.	3 days	Wed 05/06/24	Fri 07/06/24	7,8	

Project: Project2.mpp
Date: Wed 27/03/24

Page 1

ID		Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names
10			Statistical framework is provided on Modeling for surface process of weather	2 days	Mon 10/06/24	Tue 11/06/24	9	
11			Probabilistic forecasting	12 days	Wed 12/06/24	Thu 27/06/24	10	
12			Collaboration and partnerships	7 days	Fri 28/06/24	Mon 08/07/24	10,11	
13			Construct software components	2 days	Tue 09/07/24	Wed 10/07/24	12	
14			Verify and test	7 days	Thu 11/07/2	Fri 19/07/24	13	
15			Train the users and document the systems	10 days	Mon 22/07/24	Fri 02/08/24	14	
16			Install the system	2 days	Mon 22/07/2	Tue 23/07/2	13,14	




May 2024
June 2024
July 2024
August 2024
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Project: Project2.mpp
Date: Wed 27/03/24
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Duration-only
Manual Summary Rollup
Manual Summary
Start-only
Finish-only
External Tasks
External Milestone
Deadline
Critical
Critical Split
Progress
Manual Progress
Slack
Page 4

Economic Feasibility Analysis

- Benefit per year= 155,000$
- Discount rate=15%
- Recurring cost per year=85,000$
- One time cost=125,000$

	Year of Project											
Feasibility	0	1	2	3	4	5	6	7	8	9	10	**Total**
Net economic benefit	0$	155,000$	155,000$	155,000$	155,000$	155,000$	155,000$	155,000$	155,000$	155,000$	155,000$	
Discount rate(15%)	1.0	0.8695	0.7561	0.6575	0.5717	0.4971	0.4323	0.3759	0.3269	0.2842	0.2471	
PV of benefits	0$	134,772.5$	117,195.5$	101,912.5$	88,613.5$	77,050.5$	67,006.5$	58,264.5$	50,669.5$	44,051$	38,300.5	
NPV of all benefits	0$	134,772.5$	251,968$	353,880.5$	442,494$	519,544.5$	586,551$	644,815.5$	695,485$	739,536$	777,836.5$	5,146,883.5$
One-time costs	125,000$											
Net economc cost	0$	85,000$	85,000$	85,000$	85,000$	85,000$	85,000$	85,000$	85,000$	85,000$	85,000$	
Discount Rate(15%)	1.0	0.8695	0.7561	0.6575	0.5717	0.4971	0.4323	0.3759	0.3269	0.2842	0.2471	
PV of cost	0$	73,907.5$	64,268.5$	55,887.5$	48,594.5$	42,253.5$	36,745.5$	31,951.5$	27,786.5$	24,157$	21,003.5$	
NPV of all cost	125,000$	198,907.5$	263,176$	319,063.5$	367,658$	409,911.5$	446,657$	478,608.5$	506,395$	530,552$	551,555.5$	4,197,484.5$
Overall NPV												949,399$
Overall ROI												0.22618

Break-even Analysis

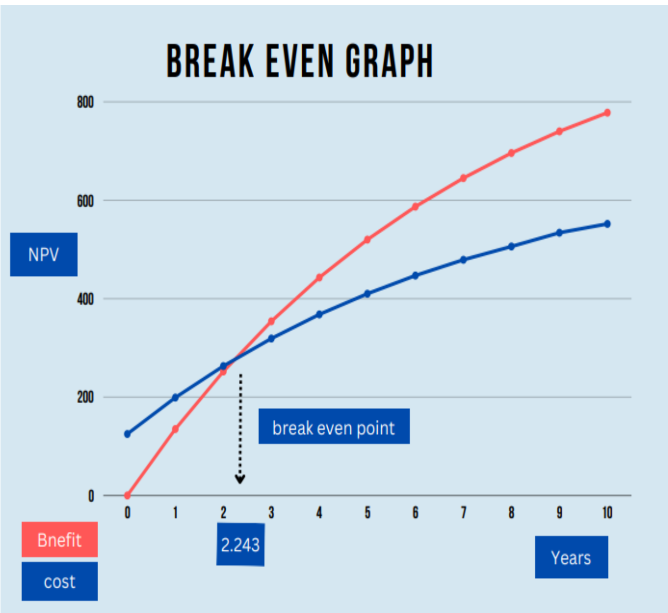
	Year of Project										
	0	1	2	3	4	5	6	7	8	9	10
Yearly PV cash flow	125,000$(-)	60,865 $	52,927$	46,025 $	40,019$	34,797$	30,261$	26,313$	22,883$	19,894 $	17,297$
Over all NPV cash flow	125,000$(-)	64,135 $(-)	11,208$(-)	34,817$(+)	74,836$	109,633$	139,894 $	166,207$	189,090 $	208,984$	226,281$

Break even Point occurs between year 2 and year 3

Break even Point =((49,025-34,817)/49,025)=0.243

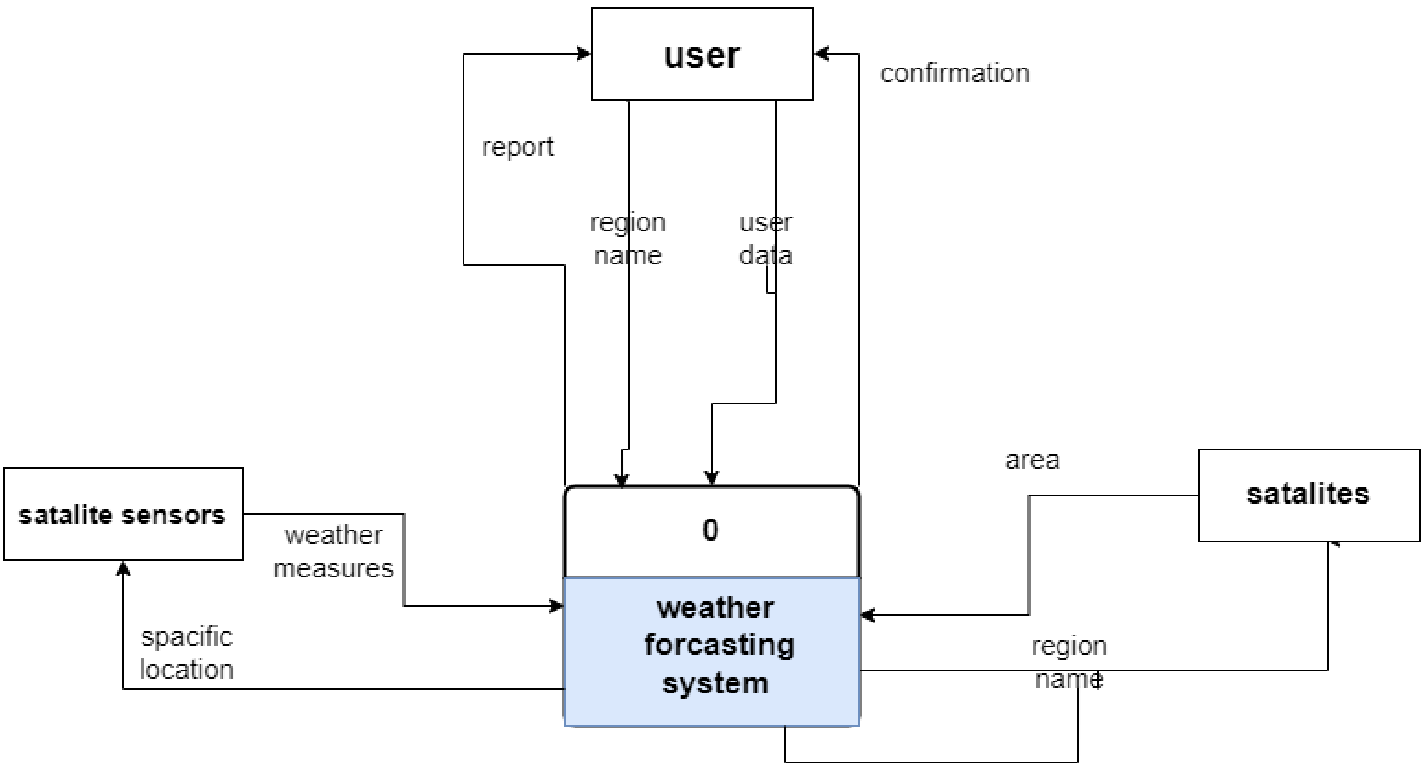
Break even Point occurs at **2.243**

Break even graph



Years	NPV Benefits	NPV Cost
0	0$	125,000$
1	134,772.5$	198,907.5$
2	251,968$	263,176$
3	353,880.5$	319,063.5$
4	442,494$	367,658$
5	519,544.5$	409,911.5$
6	586,551$	446,657$
7	644,815.5$	478,608.5$
8	695,485$	506,395$
9	739,536$	530,552$
10	777,836.5$	551,555.5$

Context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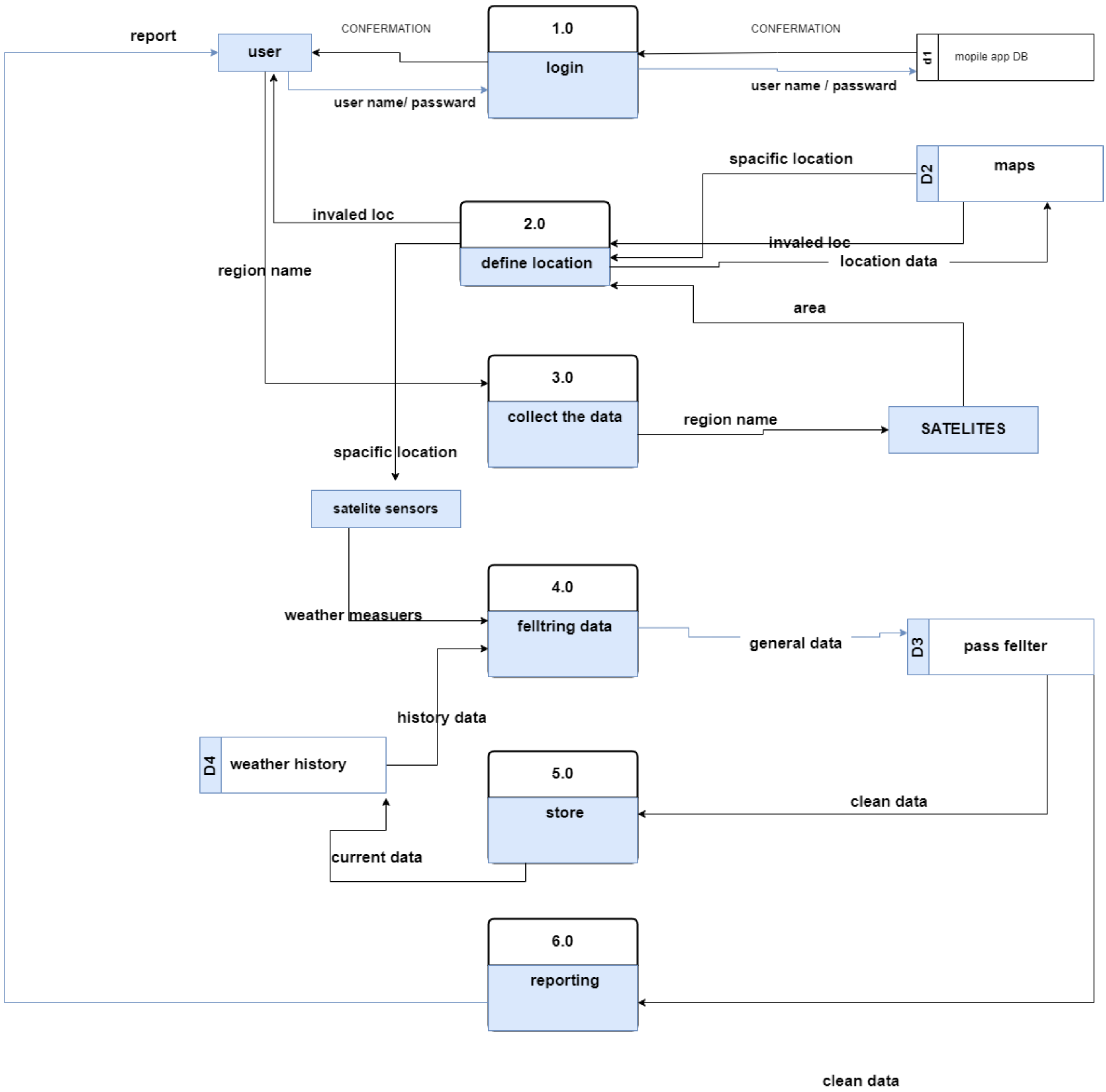
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D4
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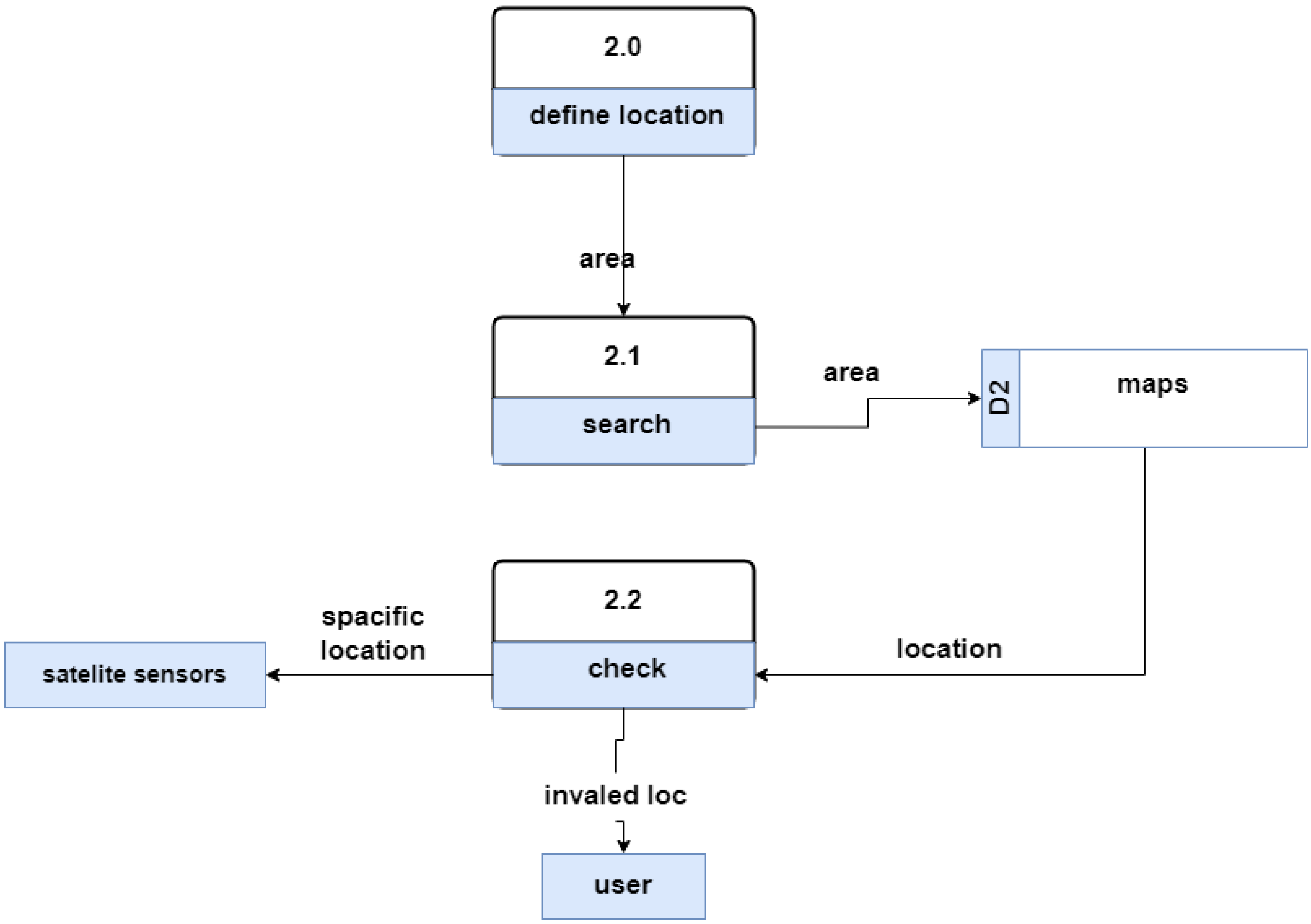
6.0
reporting
clean data

Level 1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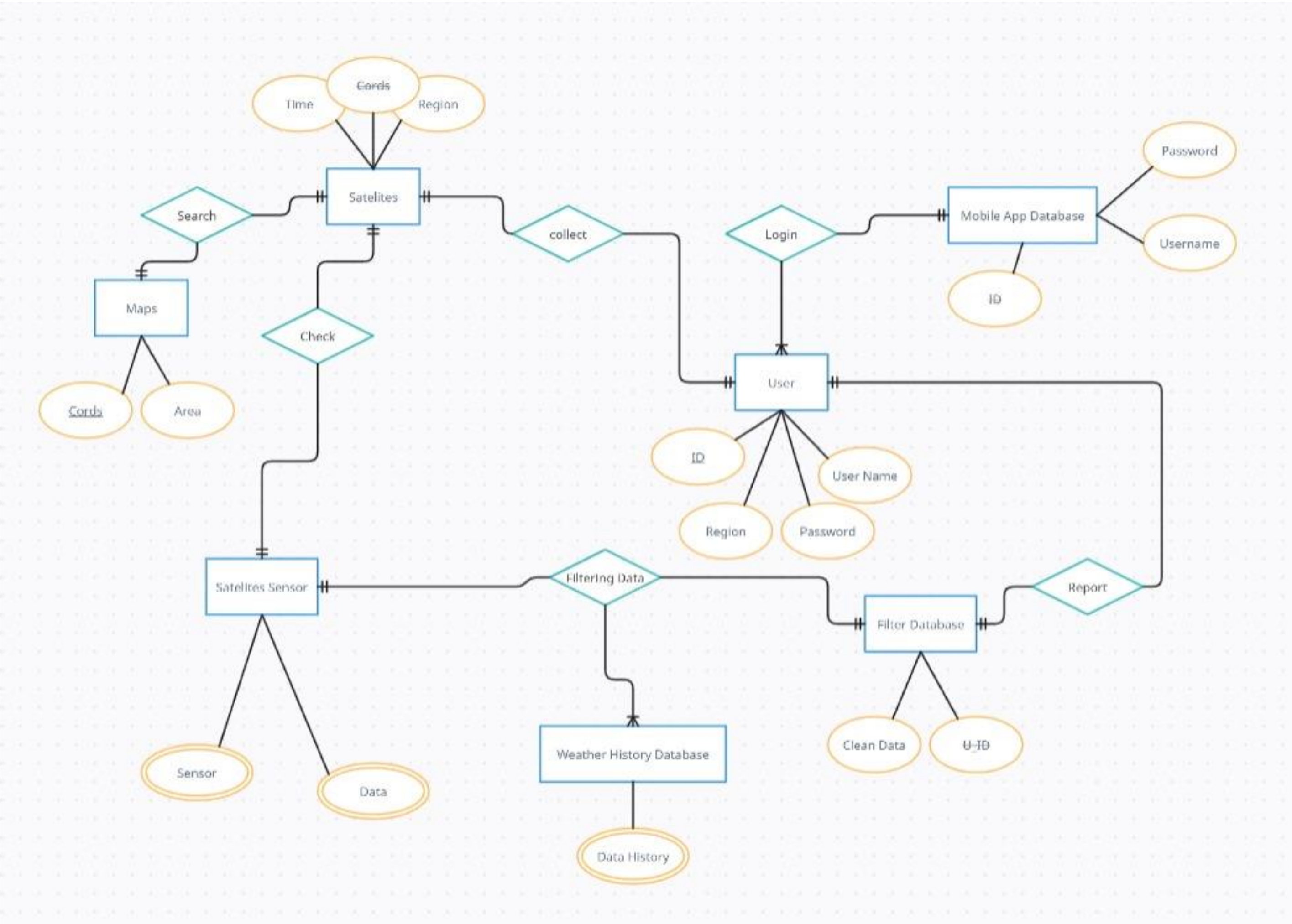
Time
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ID
Maps
Check
User
Cords
Area
ID
User Name
Region
Password
Satelites Sensor
Filtering Data
Report
Filter Database
Sensor
Data
Weather History Database
Clean Data
U-ID
Data History

Mapping

USER

ID	username	password	region	U_ID

Mobile App Database

ID	username	password

Satellites

Cords	Time	Region

Maps

Cords	Area

Satellites Sensor

Sensor Data

Sensor

Sensor	Temperature sensor	Wind speed sensor	Air humidity sensor	Rain sensor	**ID_Report**

Data

Data	Temperature	Wind speed	Air humidity	Rain

Weather History Database

ID_Report

Weather History

ID_Report	Temperature	Wind speed	Air humidity	Rain

Filter Database

U_ID	CleanData